

Nirav's EE Portfolio



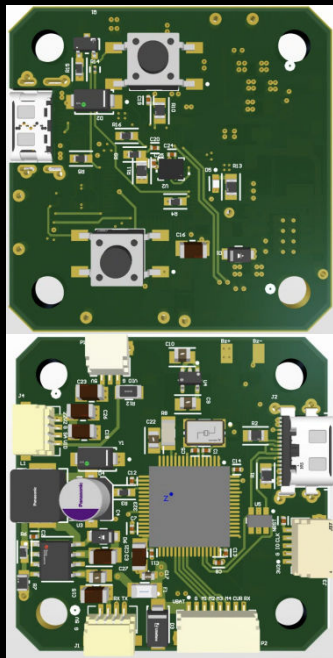
Contact me:

nirav.michelsen@gmail.com | (949) 307-2374 | [linkedin.com/in/nirav-michelsen](https://www.linkedin.com/in/nirav-michelsen) | github.com/nirav1011

Custom 5" Drone + Flight Controller*



First time flying



3d model of PCB

- **Goal:** Build and fly a 5" drone with a commercial flight controller, then design and fabricate a flight controller PCB from scratch, benchmarking it against the commercial board.

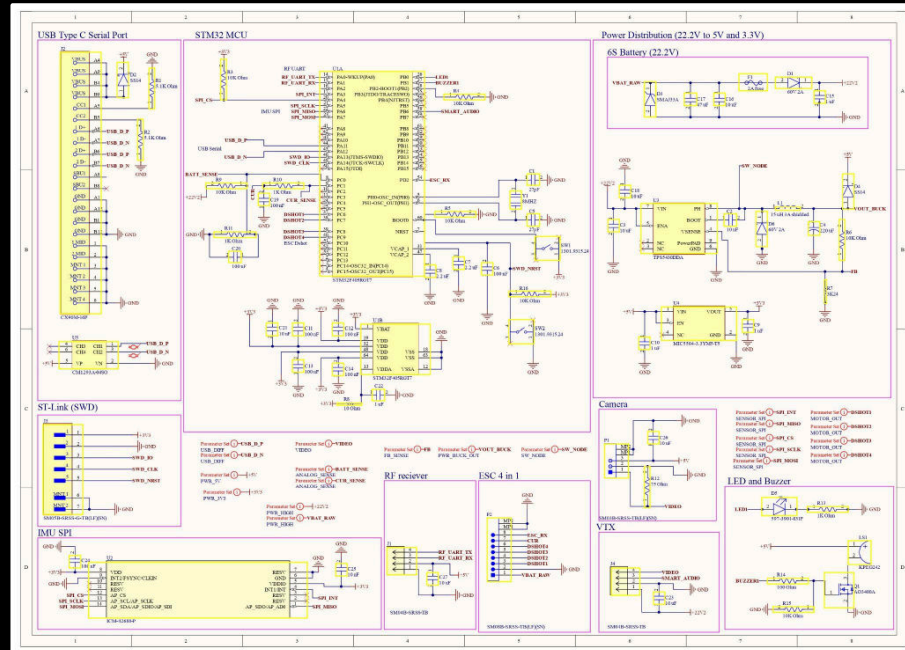
Airframe:

- 5" quad, 1750 KV 2207 motors, HGLRC 60 A 4-in-1 ESC and F405 FC

Flight controller:

- MCU: STM32F405, 168 MHz
- IMU: ICM-42688-P on SPI
- Input: 6S (22.2 V)
- Rails: 5 V buck, 3.3 V LDO
- Outputs: 4× DShot ESC, ELRS/CRSF receiver on UART
- Board: 4 layer, designed in Altium

Architecture

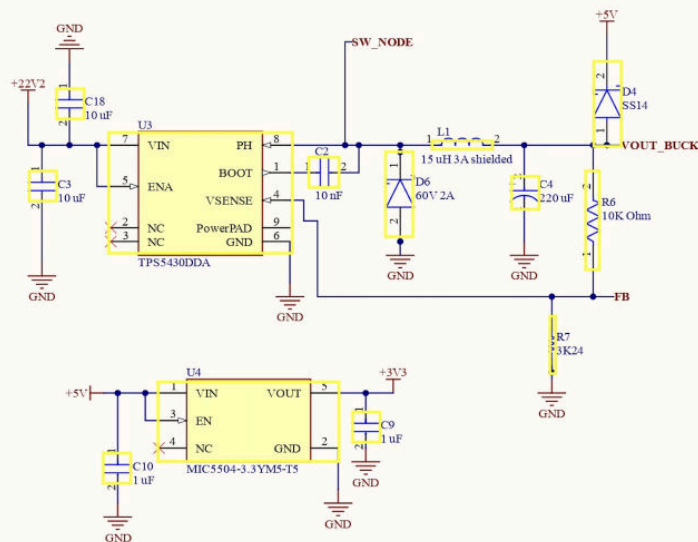
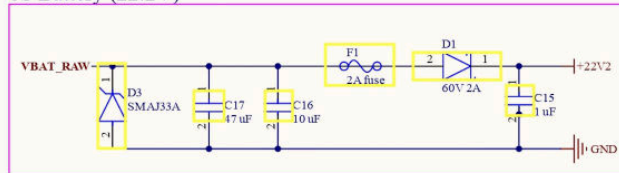


V1 schematic in Altium

- STM32F405 with the IMU on SPI, chosen for a high gyro sample rate
- Four DShot outputs mapped to one timer bank (TIM8, PC6–PC9) so all four motors update from a single synchronised source
- ELRS receiver on UART using CRSF — no inverter needed, unlike SBUS
- USB Type-C for configuration, SWD header for debugging
- LED and buzzer drivers for status and lost-model alarm

Power Distribution (22.2V to 5V and 3.3V)

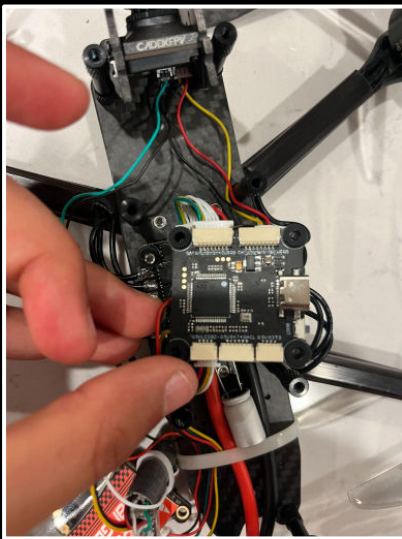
6S Battery (22.2V)



Power Tree

- TPS5430 buck converts 22.2 V to 5 V; MIC5504 LDO drops 5 V to 3.3 V
- Reverse-polarity protection on the battery input — one backwards plug otherwise destroys the board
- Schottky OR-ing between USB VBUS and the buck rail so the board can run from USB on the bench without back-feeding the battery
- Catch diode on the buck, decoupling sized to hold the MCU stable through motor current transients

Airframe and Integration



HGLRC F405 FC



Half-built drone

- Chose ELRS receiver speaking CRSF so I wouldn't need a UART inverter on the FC
- Analog camera/goggles to stay under budget and ultra-low, constant latency
- Soldered ESC, VTX and ELRS receiver to FC

Bruin Baja Telemetry System*



Our Baja car competing in AZ

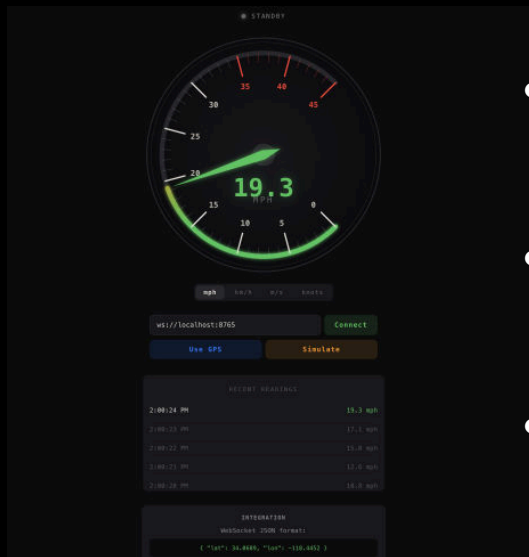
- **Context:** Bruin Racing Baja SAE builds a single-seat off-road race car for collegiate competition. The team had no way to see vehicle data during a run.
- **My role:** Electronics Hardware Project Engineer — GPS and IMU drivers, wireless link, base-station receiver, driver dashboard. (Dec 2025 – Present)
- **Outcome:** The team's first real-time telemetry capability.

GPS and IMU

- Wrote GPS and IMU device drivers in C and integrated them into the car's data acquisition system
- Logs position, orientation and acceleration continuously through a test session
- Handles the unglamorous parts: partial frames, checksum failures, malformed NMEA sentences, GPS dropout under tree cover

Enclosure of the DAQ

Antenna and Radio Link

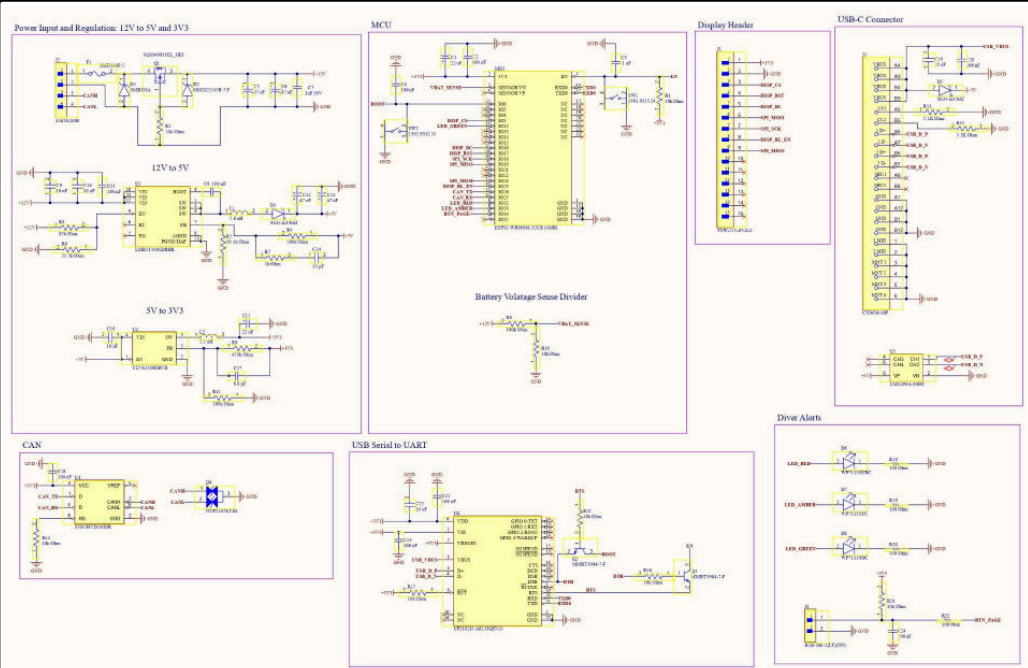


- Worked on a wireless antenna and Raspberry Pi link streaming live sensor data from the moving car to a base station
- Before this, data could only be pulled off the car after a run ended — the team was debugging blind
- **Now:** position and speed visible live during testing, so a bad run can be called in the moment

Antenna mounted to car

Local Dashboard

Dashboard Schematic*



- **ESP32-WROOM-32UE** dashboard reading live vehicle data over CAN, driving an SPI display, alert LEDs, and a page button
- **Automotive 12V front end:** PPTC fuse, SMBJ18A TVS for load dump, and a P-FET reverse-polarity block with zener-clamped gate
- **Programmable UVLO** set by the buck's enable divider (82 k/21.5 k) — clean shutdown instead of brownout when pack voltage sags
- **Two-stage switching supply** — LMR51440 (12→5 V) and TLV62569 (5→3.3 V) — with feedforward compensation on both networks
- **USB-UART** with DTR/RTS auto-reset — reflash over USB-C without opening the enclosure.

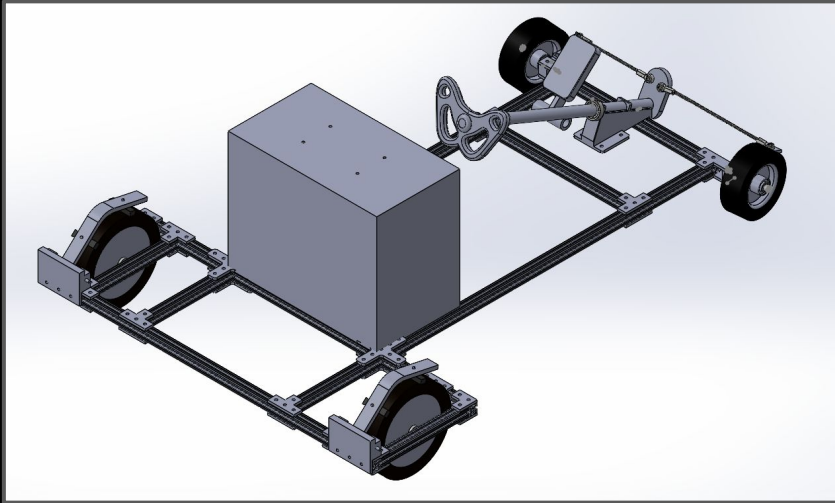
Custom Drifting Go-Kart



Me driving the Go-Kart

- **Goal:** Design and fabricate a custom battery-powered drivable go kart within a rigorous 10-weeks
- **Specifications:**
 - Top speed: 18mph
 - Power train: 36V battery, 288Wh capacity
 - Dimensions: 40in x 50in
 - Weight: 70lbs
 - Budget: Used only pre-existing components

Design



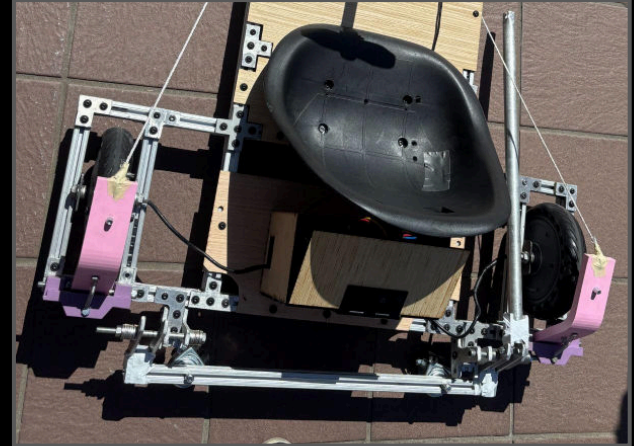
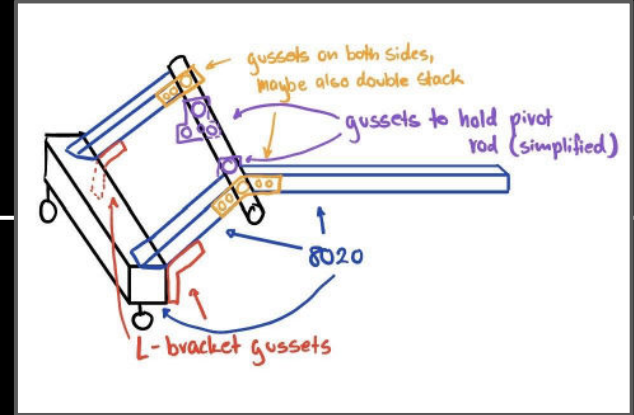
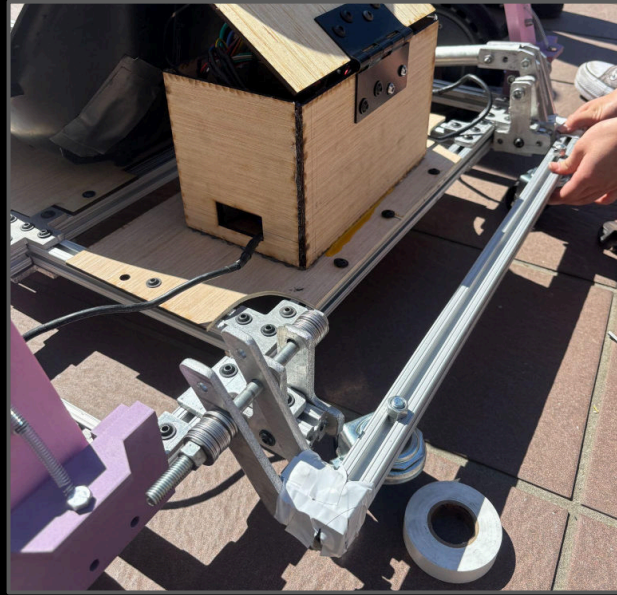
Solidworks Chassis Assembly (box is the driver seat)

- Modeled chassis, wood frame, steering column, steering wheel, and 3d printed brakes in Solidworks
- Calculated Ackerman steering geometry to minimize scrub and improve cornering
- Architected a battery-powered dual-rear drive powertrain

Castor Wheel Drifting

Idea: Aluminum handbrake-style lever to shifts the rear weight onto casters wheels, enabling omnidirectional movement

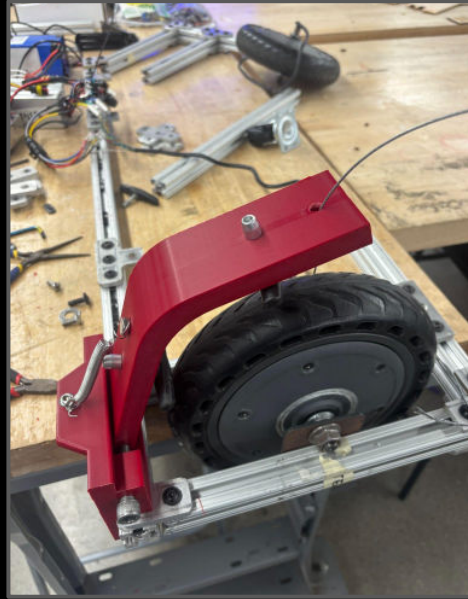
Result: Imperfect geometry and structural failures prevented full drive-wheel isolation → partial drift



Manufacturing



Cutting steel tie rods with a circular saw



Testing rear wheels and 3D printed brakes

- Laser Cutting (wood body)
- 3D Printing (brakes, steering column)
- Plasma Cutting (Gussets)
- Grinder/bandsaw on t-slotted aluminum frame

Teaching E1 Go-Karts Summer '26



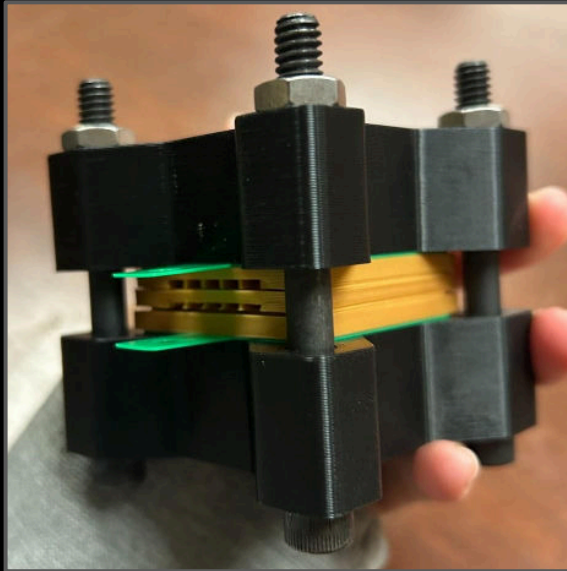
Karts and kids for the 1st cohort



All 4 summer instructors

- Designed a 3-week, 6-hour-per-day engineering curriculum from scratch for high school students with no prior experience
- Covered steering, braking, electronics fundamentals, and SolidWorks CAD
- Co-taught two cohorts
- 100% of builds completed and ran, top speed of 23 mph

Hydrogen Fuel Cell Prototype



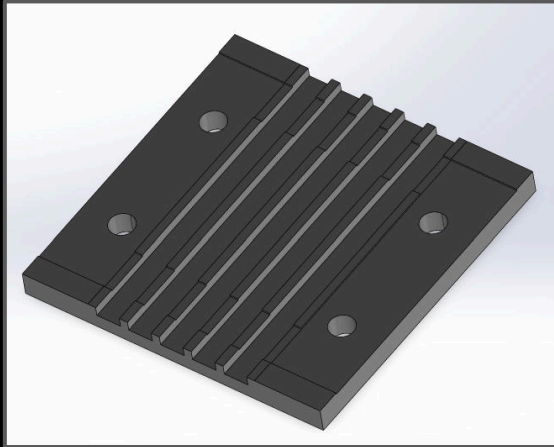
End Goal: Replace the commercial Horizon fuel cell in the Bruin Supermileage racing car with a custom stack

Target Specifications: 500W cruising power, 50% peak efficiency, voltage range between 50V-60V.

Pros of a custom stack:

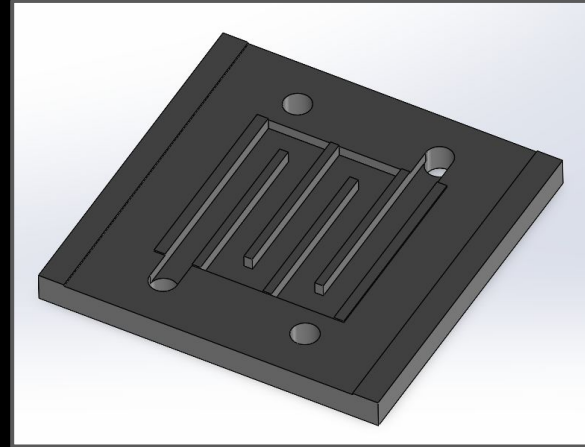
- Lightweight composites to reduce weight
- Integrate directly into chassis
- Extremely customizable (better aspect ratio)

Bipolar Plate Design



Bipolar Plate - Cathode (Parallel Channels)

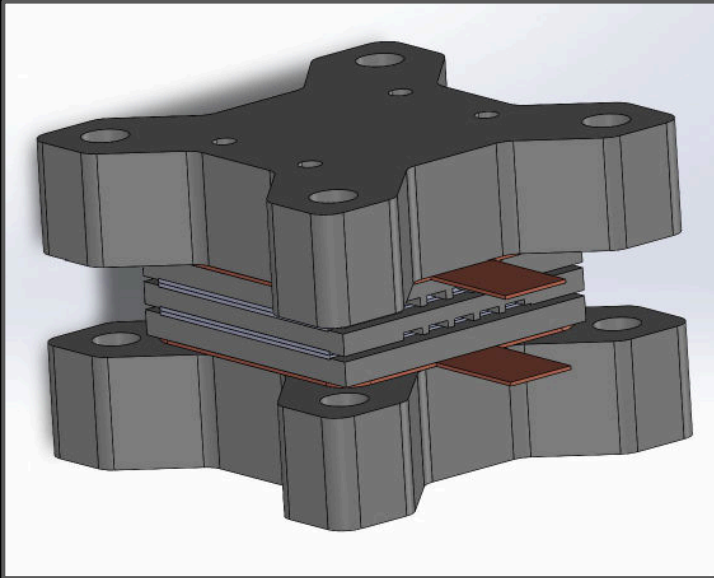
- Open channels (oxygen from the air)
- Pros: Easy to manufacture, low pressure drop
- Cons: poor water management/flow distribution



Bipolar Plate - Anode (Single Serpentine)

- Closed channels (compressed H₂)
- Pros: Good water management/flow distribution
- Cons: reactant distribution, manufacturing difficulty

Assembly



Solidworks Hydrogen Fuel Cell Stack (two MEAs)

End plate

Current collector

Cathode

Gasket - MEA - Gasket

Double-sided plate

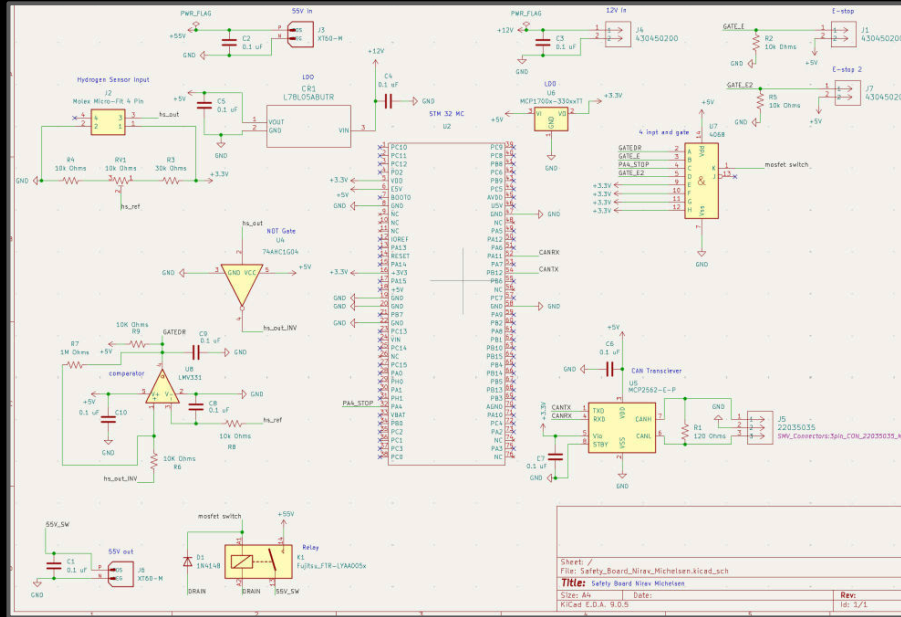
Gasket - MEA - Gasket

Anode

Current Collector

End Plate

Safety Board

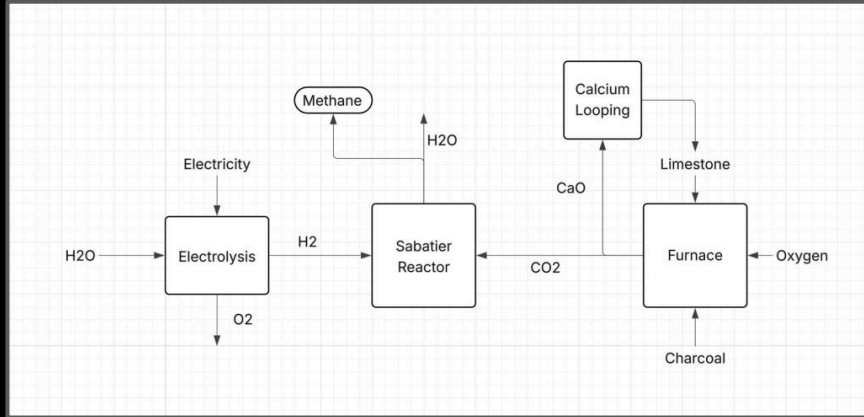


Safety Board Kicad PCB

- **Hardware Interlock:** 4-input AND gate guarantees a failsafe shutdown if E-stops, sensors, or MCU fault.
- **Hydrogen Detection:** Monitors hydrogen levels against a hardware reference for instantaneous fault triggering.
- **Power Isolation & CAN:** MOSFET-driven relay severs the 55V powertrain loop; MCP2562 CAN transceiver broadcasts vehicle telemetry.
- **Power Distribution:** Multi-stage LDO system steps 12V down to stable 5V/3.3V rails

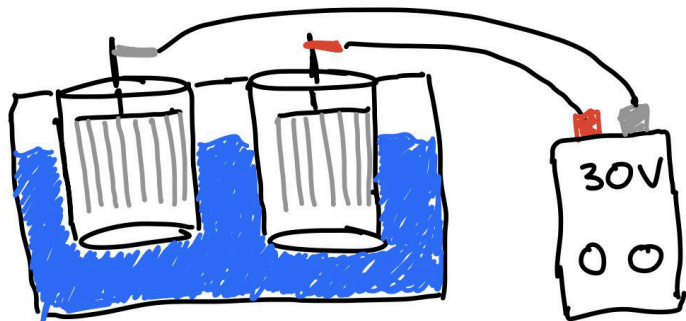
Carbon Capture Prototype

Goal: synthesize natural gas using atmospheric CO₂



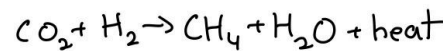
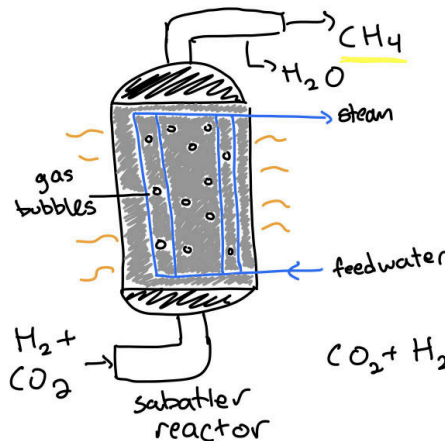
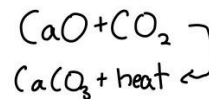
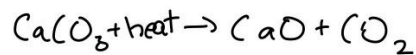
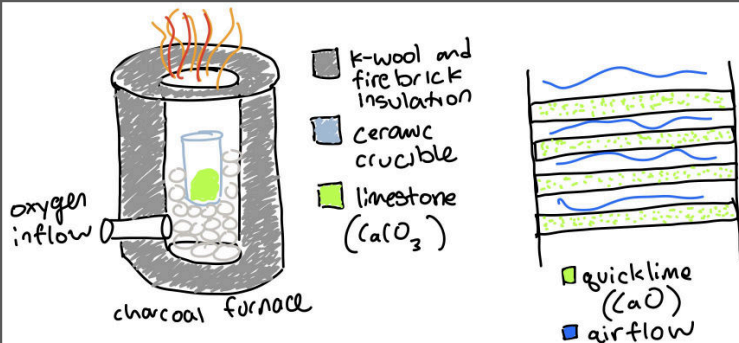
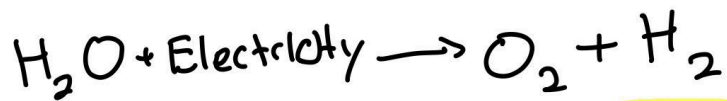
Process Flow Diagram of entire process

- **Hydrogen Generation:** Utilizes an Electrolyzer to split water into high-purity Hydrogen and Oxygen
- **Carbon Sourcing:** Calcinate limestone to create concentrated CO₂ and quicklime to recarbonate atmospherically.
- **Methane Synthesis:** Hydrogen and Carbon Dioxide into a Sabatier Reactor with a nickel catalyst.
- **Applications:** Create a fully renewable and high energy density fuel compatible with existing natural gas infrastructure (if powered by solar).

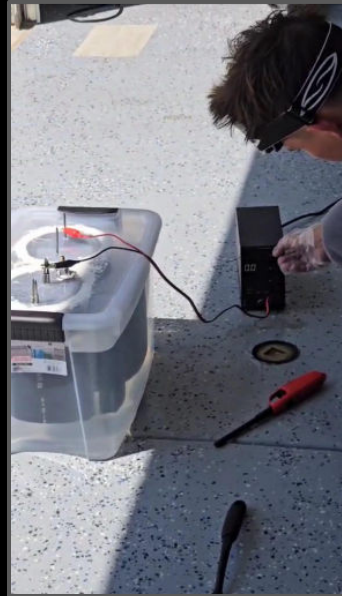


Electrolyzer

- stainless steel electrodes
- distilled H_2O with potassium hydroxide as electrolyte



Design and Manufacturing



Electrolysis



Electrolyzer Cathode/Anode
design (with only 2 steel sheets)

Improvements

Did **not** sustain methane production

Main Issues:

1. Moisture remnants in the Hydrogen gas
2. Inefficient Electrolysis due to gas shielding
3. Furnace could not pass 900°F
4. Catalyst poisoning due to sulfur from charcoal

Improvements:

1. Gas-liquid separation
2. Metal mesh electrodes for more uniform current density
3. Increase oxygen airflow (larger hole, use mattress pump instead of hair dryer)
4. Use of electric, solar-powered furnace instead of charcoal

Hobbies

I am also passionate about sports (especially soccer), the gym, nature, hiking, camping, and am learning to play the guitar!



Contact me: